

ELI LILLY AND COMPANY
HEALTH AND SAFETY PROCEDURE
North American Logistics Operations

Electrical Safety Program

USD-SOP-Electrical Safety Program-013

PURPOSE	To prevent injury to all personnel who work with or near electrical hazards by promoting an electrically safe workplace free from unauthorized exposure to electrical hazards for all employees and contractors. Utilization of this program as a basis for the site electrical safety program, as outlined in the Occupational Health and Safety Policy, will help ensure that the site program is in compliance with applicable regulations. Where local requirements differ from these requirements and do not compromise safety, modifications should be made to reflect these differences. This program is based on the references listed in this document. All applicable elements of these references are part of this program. The items specifically addressed in this program are either to highlight areas of special concern to Eli Lilly or to address areas not mentioned or subject to interpretation in the aforementioned documents. Where differences are inconsequential, modification is appropriate.
AREAS INVOLVED	North American Logistics Operations (NALO) Supervision and Management (includes Quality Assurance), and NALO Engineering and Maintenance
REASONS FOR REVISION	All revisions are marked with an (*): - To add USD-SOP-NALO Equip Lockout and Line Breaking Pgm-001 to the References section. - Rationale: It is mentioned in the Safety Related Work Practices section. - To change "Alliance Contractor" to "Qualified Contractor". - Rationale: To generalize the term, "contractors". - To add air fryers to the Installation and Operation Requirements section. - Rationale: As an example of a prohibited certain type of permanently installed heat producing appliance. - To add "EH rated" to "safety shoes" in the Safety Related Work Practices section. - Rationale: To describe what all QEWs must wear.
APPROVALS	Approval of this procedure requires the signature of: Representative, Health, Safety, and Environmental Management, NALO Operations Management Management, NALO Quality Assurance

REFERENCES

DOHS067: Electrical Safety Awareness in NALO
NFPA 70E, Standard for Electrical Safety in the Workplace
OSHA-29 CFR 1910. 137,29 CFR 1910.269 and 29CFR 1910.301-399
EN 50110-1: 1997 Operation of electrical installation
Occupational Health and Safety Policy
Confined Space Entry Policy
Plant Electrical Safety Policy
Equipment Lockout Policy
Fall Protection Policy
Corporate Electrical Safety Policy
Corporate Electrical Safety Requirements
GPME-SOP-SAF-IN-FIRE-1004-179, Fire Safety - Appendix A
*USD-SOP-NALO Equip Lockout and Line Breaking Pgm-001

**TABLE OF
CONTENTS**

Topic	Page Number
Scope	4
Definitions and Terms	6
Installation and Operation Requirements	9
Safety Related Work Practices	17

-
- SCOPE** The Electrical Safety program is a minimum standard and applies to all manufacturing, research and development, sales and marketing, and distribution sites. The scope of implementation may vary depending on the functionality and size of a site. The responsibilities listed may be combined or otherwise be accomplished as determined by each site. The US distribution centers conform to the Corporation's electrical safety requirements, and to the local, and country jurisdiction. When there are differences between regulations, the more stringent must be followed.
- A. NALO Electrical Safety Committee (comprised of the Engineering & Maintenance group)
1. Develops and maintains an Electrical Safety Program in agreement with the Corporate Electrical Safety Policy, the Corporate Electrical Safety Requirements and all applicable laws and codes
 2. Develops self assessment program
 3. Reviews Electrical Safety aspects of proposed facility changes
 4. Recommends priorities for evaluating and controlling potential electrical hazards
 5. Reviews electrical installation, operation, and maintenance procedures
 6. Utilizes CAPA Change Management as a process to inspect facility equipment upon installation/changes for compliance with Corporate requirements, (US)NEC, and OSHA 29 CFR 1910 Subpart S
 7. Reviews electrical accidents and incidents
- B. Facility Operation and Maintenance Management
1. Ensures that employees are provided a workplace that is free from recognized hazards
 2. Ensures that employees performing electrical work are trained and qualified
 3. Ensures that approved, maintained, and tested personal protective equipment and clothing is provided, available, and used properly
 4. Establishes, implements, and maintains procedures and practices that will ensure safe conduct of electrical work
 5. Keeps and maintains records as required
 6. Ensures management sponsorship, support and participation in the Electrical Safety committee
 7. Ensures that all contractors performing electrical work at the facility are qualified and adhere to electrical safety requirements
 8. For facilities that have contracted out electrical maintenance, ensures that the contractor has an acceptable Electrical Safety Program.
- C. Facility Engineering and Construction Management
- Implements the Electrical Safety Program by ensuring that the design, installation, and modifications of facilities are in compliance with OSHA and NFPA guidelines (US) and by ensuring that all of the engineers and designers receive proper training.
-

Continued on next page

**SCOPE,
CONTINUED**

- D. NALO Safety Representatives
 - 1. Assist management and the electrical safety committee in implementing the Electrical Safety Program
 - 2. Are members of Site Electrical Safety committee
 - 3. Assist in maintaining statistical records
 - 4. Identify, evaluate, and provide notification of hazardous conditions
 - 5. Investigate and report all electrical accidents, incidents, and near misses.
 - E. NALO Management
 - Implement the Electrical Safety Program.
 - F. Management and Supervision of Electrically Qualified personnel
 - 1. Implement the Electrical Safety Program by developing electrically qualified personnel through appropriate training
 - 2. Assign only personnel qualified for a task, to the task
 - 3. Keep records of equipment testing
 - 4. Maintain records of switching procedures as necessary (the plant site may elect to have this responsibility with the Plant EE or EE technician)
 - 5. Ensure that personnel receive task-specific electrical safety training as required by OSHA 29 CFR 1910.332 and Electrical Safety Awareness Training (US).
 - 6. Ensure that approved, maintained, and tested personal protective equipment and clothing is provided, available, and used properly.
 - G. Employees and *Qualified Contractors
 - 1. Employees/contractors are responsible to comply with occupational safety and health regulations and standards that apply to their own actions and conduct
 - 2. Employees/contractors are responsible for the immediate reporting to management of unsafe and unhealthful conditions.
 - H. Other than Electrically Qualified Personnel
(see the Special Equipment Section, B, 2)
 - 1. Must have an awareness of the electrical hazards in their workplaces
 - 2. Report electrical shocks, failures, and discovered hazards
 - 3. Attend appropriate electrical safety awareness training.
 - I. *Qualified Contractor Management
 - 1. Maintain records of qualifications and training in personnel files
 - 2. Provides training on Lilly Electrical Requirements and local regulations.
 - J. Plant Site Human Resources
 - Maintain records of training and qualifications in personnel files.
-

**DEFINITIONS
AND TERMS**

Adapter cord (Lilly): A short, 1 to 3 foot extension cord with an explosion proof plug and a general use receptacle used to power general use equipment in a Hazardous (Classified) area.

Approach boundary (NFPA 70E) :

Boundary Type	Description
Arc Flash	When an arc flash hazard exists, an approach limit at a distance from a prospective arc source within which a person could receive a second-degree burn if an electrical arc flash were to occur. (Personnel must wear proper personal protective equipment within this boundary during periods of energized work.) If the actual arc flash boundary distance has not be posted or determined, it will be assumed to be at a minimum of 4 feet for systems operating up to 1000 volts and at a minimum of 20 feet for systems operating above 1000 volts.
Limited Approach	An approach limit at a distance from an exposed energized electrical conductor or circuit part within which a shock hazard exists.
Restricted Approach	An approach limit at a distance from an energized electrical conductor or circuit part within which there is an increased likelihood of electric shock, due to electrical arc-over combined with inadvertent movement.

Approved (NEC): Acceptable to the Authority Having Jurisdiction (AHJ)

Authority having jurisdiction (AHJ)(NEC): Interprets the requirements of the National Electrical Code (NFPA70); The National Electrical Safety Code (ANSI C2); 29CFR 1910, Subpart S; 29 CFR 1926, subparts K and V; and this document.

Double Insulation: An insulation system where two separate layers of nonconductive materials are placed around current carrying components of tools to prevent leakage of current. **Note:** Not effective for appliances that are immersed in water or soaked in rain.

Electrical Shock, Reportable (Lilly): Any electrical shock (other than static) is classified as an incident and must be reported immediately to health services and supervision. The employee must not attempt to evaluate the severity of the shock or its effects without medical consultation.

Electrically Safe Work Condition (NFPA 70E): A state in which the conductor or circuit part to be worked on or near has been disconnected from energized parts, locked/tagged in accordance with established standards, tested to verify the absence of voltage, and if necessary, temporarily grounded for personnel protection.

Energized Exposed Part (OSHA): An energized part that can be inadvertently touched or approached nearer than a safe distance.

Energized Electrical Work (OSHA): Work performed on a component of an electrical power system within the reaching, stumbling, or falling distance of energized exposed parts.

Continued on next page

**DEFINITIONS
AND TERMS,
CONTINUED**

Entry (Lilly): Any action resulting in any part of the body breaking the plane of any opening of an electrical underground enclosure.

Examination (OSHA): Examination process described in 29 CFR 1910.303(b)(1), "Examination" and NFPA 70, Article 110.3, "Examination, Identification, Installation and Use of Equipment". These examinations are performed by a qualified electrical worker to ensure that electrical equipment is free from recognized hazards that are likely to cause physical harm.

Exposed Yard Work (Lilly): Work in outdoor substations which involves conductors in an energized state, e.g. manual switching. Does not include inspections.

Extension Cord: Wiring used to lengthen a service cord. Temporary use only (30 days or less).

Hot Work/Fire Hazard Permit: A permit that is discussed and followed to manage the risks associated with any activity involving burning, welding or similar operations that are capable of initiating fires or explosions. See Gensuite site for permit.

Ground-Fault Circuit-Interrupter (GFCI): A device intended for the protection of personnel that functions to de-energize a circuit or portion thereof within an established period of time when a current to ground exceeds some predetermined value that is less than that required to operate the overcurrent protective device of the supply circuit.

Integrated Facilities Management Provider (IFMP) Employees

Also known as CBRE employees.

Labeled (NEC): Equipment or materials to which a label, symbol, or other identifying mark has been applied by a Nationally Recognized Testing Laboratory (NRTL).

Listed (NEC): Equipment or materials included in a list published by a NRTL.

Long Term Work Entry (Lilly): Entry into an electrical underground enclosure for construction, cleaning, or repair of equipment within the underground enclosure.

Nationally Recognized Testing Laboratory (NRTL) (OSHA): An organization acceptable to the AHJ and concerned with product evaluation that maintains periodic inspection of production of listed equipment and materials. Refer to 29 CFR 1910.7 for further information.

Personnel (Lilly): Employees and on-site contractors.

Qualified Electrical Worker (QEW) (Lilly):

One who has demonstrated skills and knowledge related to the construction and operation of electrical equipment and installations and has received safety training to identify the hazards and reduce the associated risk. Qualifications must be documented. Known as Qualified Person (NFPA 70E).

Continued on next page

**DEFINITIONS
AND TERMS,
CONTINUED**

Relocatable Power Tap (RPT) (UL): A device that consists of a grounded attachment plug and a flexible power cord terminated in an enclosure in which one or more receptacles are mounted. Also known as a power strip, strip plug, surge protector, or surge strip

Safe Approach Distances: Flash protection boundaries that are derived from specific Voltage ranges. See NFPA 70E for information

Service Cord: Wiring to supply power to electrical equipment. Cord is permanently connected to the equipment and has a plug on one end.

Short Term Observation Entry (Lilly): Entry into an electrical manhole that includes all interior enclosure observations required for job planning, finding faults, cable identification, etc.

Standby Power System: A second source of power, usually, but not always connected through an Automatic Transfer Switch. Could be another circuit, generator, UPS, or battery station.

Temporary Wiring: Wiring method, such as an extension cord, used to replace permanent building wiring for a short period of time. Temporary wiring shall be permitted during the period of construction, remodeling, maintenance, repair, or demolition of buildings structures, equipment or similar activities. Temporary wiring shall be permitted during emergencies, tests, experiments, and developmental work. There is no time limit on these activities, but the wiring will be removed immediately upon the completion of the purpose for which it was installed. There shall be a 90 day limit for temporary wiring supplying holiday decorations and lighting

Unqualified Affected Person (Lilly)

NFPA 70E defines Unqualified Persons. Lilly creates the definition of Unqualified Affected Person, as anyone who is not skilled, trained and familiar, or equipped, to perform general or specific electrical tasks shall be trained in, and be familiar with, any electrical safety-related practices necessary for their safety. Such persons, working in areas where electrical safety is a concern.

Personnel not trained as a QEWS who may work with QEWS.

Visual Inspection (Lilly): Entry into an electrical manhole in which an actual descent is NOT made, but the seal of the electrical manhole lid is broken or removed and other actions may occur resulting in any part of the head or feet breaking the plane of the manhole.

**INSTALLATION
AND
OPERATION
REQUIREMENTS**

- A. Clearance for Workspace in Front of Panels
Reference: See NEC Art. 110.26.
- B. Extension Cords
1. When portable tools are used with an extension cord, a GFCI must be used. This requirement is not effective in areas where ground fault sensing breakers are installed in panels as part of a normal installation. Test equipment and procedures that might be affected by nuisance trips can be exempted from this requirement.
 2. No extension cords are used on a permanent basis. They may be used as temporary wiring in compliance with NEC article 590.
 3. All extension cords must be UL Listed.
- C. Adapter Cords
1. Adapter cords are not used without a hot work/fire hazard permit and the maximum length for an adapter cord is three feet.
 2. Adapter cords are owned and controlled by the management of the hazardous area involved.
 3. Adapter cords are checked annually for ground continuity with checks recorded.
- D. Flexible Cords
1. Flexible cords may be used in the following situations:
 - a. Pendants (must have strain relief)
 - b. Light fixtures
 - c. connection of portable lamps, signs, or appliances
 - d. elevator cables
 - e. wiring on cranes or hoists
 - f. connection of utilization equipment to facilitate frequent interchange (including shop equipment that is frequently moved)
 - g. prevention of the transmission of noise or vibration
 - h. appliances where the fastening means and mechanical connections are specifically designed to permit ready removal for maintenance or repair, and the appliance is intended or identified for flexible cord connection
 - i. data processing cables
 - j. connection of moving parts
 - k. temporary wiring
 - l. manufacturer installed flexible cords

Reference: See NEC Art. 400.10 Uses Permitted.

Continued on next page

**INSTALLATION
AND
OPERATION
REQUIREMENTS,
CONTINUED**

2. Flexible chords may not be used in the following situations:
 - a. Substitute for fixed wiring of a structure
 - b. run through holes in walls, structural ceilings, suspended ceilings, dropped ceilings, or floors
 - c. run through doorways, windows, or similar openings
 - d. attached to building surfaces (except where concealed by walls, floors, or ceilings or installed in raceways, except as permitted elsewhere in the NEC
 - e. used or installed in a manner that unduly exposes them to physical damage.

Reference: See NEC Art. 400.8 Uses Not Permitted.

3. Some basic rules to always keep in mind include:
 - a. Always use a cord and fittings identified for the application
 - b. never use flexible cord or cable in place of permanent wiring method
 - c. replace (rather than repair) damaged cord or cable
- E. Relocatable Power Tap (RPT) (also known as Power Strips)
 1. RTPs are intended for use on 120v circuits and are rated at least 15 amps. They are intended for indoor and dry location uses as a relocatable multiple outlet extensions of a branch circuit for laboratory equipment and to provide outlet receptacles for computers and their associated hardware components
 2. Acceptable RTPs at Eli Lilly will be provided with a resettable circuit breaker, on/off switch, and surge suppression. Maximum length of the flexible cord will be 6 feet (unless approved by management). The RPT must be UL listed/ labeled.
 3. RTPs are plugged directly into a permanently installed branch circuit receptacle. Under no circumstance will a RPT be plugged into an extension cord or plugged into another RPT (daisy chained).
 4. Extension cords will not be plugged into an RPT.
 5. RTPs will not be "permanently" secured to building structures or furniture. Attachment hardware that comes from the RPT manufacturer may be used if the removal of the RPT requires no tools.
 6. The flexible cord of the RPT shall not be routed through walls, windows, doors, ceilings, floors or similar openings. The flexible cord of the RPT will not be routed under carpeting or a rug.
 7. RTPs may only be used in offices, laboratories, conference rooms, and workstations with computers in warehouses, production areas and shops. They are not permitted in maintenance areas of the shops, process areas, break rooms, kitchens, construction areas, hazardous locations, outdoor locations, or damp locations.

Continued on next page

**INSTALLATION
AND OPERATION
REQUIREMENTS,
CONTINUED**

8. RPTs may be used to supply:
 - a. Low power usage laboratory equipment
 - b. Computers and associated hardware peripherals
 - c. Telephones
 - d. Low wattage battery chargers
 - e. Electric desktop staplers
 - f. Electric desktop pencil sharpeners
 - g. Desktop clock/radio
 - h. Small personal use fans
 - i. Desk lamps
9. RPTs may not be used to supply:
 - a. Appliances

Examples: refrigerators, coffee pots, space heaters, microwave ovens, toasters, toaster ovens, crock-pots, electric skillets, electric grills, blenders, mixers, etc.

- b. Portable power tools
- c. Pedestal fans
- d. Copiers
- e. Laser printers
- f. High power loads (for the purpose of this paragraph is defined as any device which draws over 2.5 amps. Loads up to 5 amps, may be supplied with management approval)

F. Electrical Service Cords in Production Areas

1. The maximum length of service cords will be 6 feet for single phase and 12 feet for three phase unless otherwise specified by management of the area that owns the equipment.
2. If a long cord creates a safety or housekeeping problem, a new outlet should be provided. Flexible cords and cables shall not be used as a substitute for fixed wiring of a structure.
3. Reference: See NEC article 400.12.
4. Service cords shall be checked before each use for evidence of physical damage. Damaged cords shall be removed from service.
5. In production or service departments that require special length cords as standard equipment there must be a reel, spool, or hanger to keep the excess cord off the floor.

Under no conditions will explosion proof plugs be permanently attached to non-explosion proof tools or equipment. Adapter cords must be used in rated areas

Continued on next page

**INSTALLATION
AND
OPERATION
REQUIREMENTS,
CONTINUED****G. Electric Portable Hand Tools and Other Devices**

1. Shall be equipped with 3-prong plugs unless listed as double insulated.
2. Service cords should be checked before each use for evidence of physical damage. Damaged cords should be removed from service.
3. Portable heaters and low current radiant heat floor mats are permitted only when issued by Lilly maintenance personnel.
4. Only UL certified portable heaters with the following features will be issued:
 - a. Tip-over shutoff with manual reset
 - b. Automatic overheat shutoff
 - c. Cool-touch housing
5. Electrical devices should be inspected by the user for any damage to the cord or obvious hazards. If any damage or hazards are found the device must be removed from service immediately.

Examples:

- Electric fan
- Radio
- Pencil sharpener
- Refrigerator
- Lamp
- Microwave, and
- Coffee pot

H. Small Appliances

1. Electrical Safety is an important element of a comprehensive fire safety program. Portable electrical equipment provided by Lilly will be inspected prior to issuance.

Any portable electrical equipment that does not function properly will be removed or tagged out of service until it is repaired.

Portable electrical equipment such as small appliances shall be categorized into one of two classifications:

a. Heat Producing Portable Electrical Equipment

- 1) Heat producing electrical equipment: coffee makers, refrigerators, microwaves and toasters shall be provided by the Company at its discretion. Electrical equipment in this category may not be brought in from home by employees or contractors. The only exception to this is when crockpots are brought in with food to keep it warm. In this event, Appendix A from GPME-SOP-SAF-IN-FIRE-1004, Fire Safety must be completed, submitted, and approved prior to the event date. An electrical inspection by Maintenance must also be completed with an applied dated sticker.

Continued on next page

**INSTALLATION
AND OPERATION
REQUIREMENTS,
CONTINUED**

- 2) This type of heat producing electrical equipment shall meet the following criteria:
 - i. Confined **ONLY** to designated break areas. The building owner will have sole responsibility for deciding which spaces can be designated a break area.
 - ii. Must have a sustainable Preventive Maintenance (PM) schedule.
 - iii. Example: Must be plugged in directly to an outlet, not an RPT.
- 3) Certain types of permanently installed heat producing appliances are prohibited.

Example:

- toaster ovens/*air fryers
- George Foreman grills or like devices
- griddles
- convection ovens
- coffee cup or candle warmers

b. Non-Heat Producing Portable Electrical Equipment

- 1) Certain types of non-Lilly purchased portable electrical equipment are allowed to be brought in by employees. This equipment is defined as non-- heat producing.

Examples:

- pencil sharpeners
- staplers
- fans
- radios

- 2) This type of portable electrical equipment shall meet the following requirements:
 - i. meet UL, FM or equivalent standards
 - ii. have no apparent defects or broken /worn parts
 - iii. be properly grounded if required
 - iv. be regularly visually inspected by the user

Continued on next page

**INSTALLATION
AND OPERATION
REQUIREMENTS,
CONTINUED****2. Designated Break Area Support**

- a. Break areas designated by the company will be designed to accommodate space for heat producing appliances. The appliances will be purchased by the company and regularly inspected by the building owner.
- b. Designated break area elements:
 - 1) Common spaces open to use by multiple people. Not a personal space.
 - 2) Area behind and around the appliance is well ventilated.
 - 3) Ample supply of electrical outlets that provide each plug in cord a direct connection. No RPTs or power strips.
 - 4) Back splashes are solid surfaces such as drywall or other hard surfaces, not soft fabric covered walls such as cubicle dividers.
 - 5) If provided, counters and cabinets are of a fixed permanent nature.
 - 6) Combustibles, with the exception of minimum quantities available for immediate use such as cups, utensils or condiments, are kept in closed cabinets or stored elsewhere.
 - 7) Break area is recognized by the building engineer or owner as a permanent space.

3. Inspection

Per Indiana Fire Code, each IFM managed building is inspected floor by floor annually according to the inspection schedule maintained by the Lilly Site Fire Marshal in conjunction with the building engineer.

Observations of a noncompliant electrical appliance or heating device will result in the observation being placed on the building report. The observation will be tracked by the Lilly Site Fire Marshal in the building inspection report.

Reference: GPME-SOP-SAF-IN-FIRE-1004, *Fire Safety*.

4. Observation Report Distribution

At the conclusion of the annual building inspection, the Lilly Site Fire Marshal shall issue a report containing the date, building, floor, and cubicle number for a personal space or column number for a common space.

The report will be sent to the distribution list for the annual building inspection.

I. Ground Fault Circuit Interrupter (GFCI)

GFCIs will be used according to NEC. In addition, GFCIs will be tested at least annually per manufacturer's instructions.

Reference: See NEC Chapters 2 and 4.

Continued on next page

**INSTALLATION
AND OPERATION
REQUIREMENTS,
CONTINUED**

- J. Temporary Wiring Methods
Reference: See NEC Article 590.

- K. Electrical Identification
Methods of electrical equipment identification shall be developed for each location in compliance with local requirements. The following methods shall be used:

1. Medium Voltage Switchgear (1 to 25 kV)
Main breaker or disconnect shall be labeled as main with bus number, power source and ampere rating.

Feeder breakers shall be labeled with the breaker number, circuit number (if different), normal load ID and location, emergency/alternate load ID and location, and ampere rating.

2. Unit Substations (Load Centers)
Medium Voltage (MV) disconnect shall be labeled with the load center number, the power source and location, the MV circuit number, the ampere and nominal voltage rating.

Secondary main breaker shall be labeled as "Main", with cubicle number.

The "tie" breaker shall be labeled as "Tie", with cubicle number. Feeder breakers shall be labeled with cubicle number, load identification, and load location.

3. Motor Control Centers (MCC)
Main Breaker, disconnect, or incoming lug location shall be labeled as main, with MCC number, power source, power source location, cubicle number, and bus rating (if determined important by plant engineer). Any shunt trip locations or special equipment should also be identified.

Each cubicle (bucket) shall be labeled with its cubicle number, load identification, location of load, and any interlocks or interconnections with other equipment.

4. Isolation Switches/Combination Starters/Adjustable Speed Drives
Shall be labeled with the power source, power source location, load identification (required by code if not evident), and interlocks or interconnections with other equipment.

5. Lighting and Distribution Panels
The outside tag shall contain the panel number, the power source, power source location, and voltage level.

The inside schedule shall contain the load description and load location for each breaker and the breakers should each be labeled with a breaker number.

6. Control Panels
The outside tag shall contain the panel number, power source, power source location, voltage level, and a warning if there are mixed voltages or interlocks with other equipment. Panels installed after 1 January 2005, must indicate the short circuit current rating (SCCR).
An up-to-date wiring diagram is permitted to be in a door pocket inside.

Continued on next page

**INSTALLATION
AND OPERATION
REQUIREMENTS,
CONTINUED**

7. Standby power sources
Whenever a standby source is connected to the system, a sign shall be placed at the equipment where the connection is made at the load indicating the type and location of the standby source.
8. Distribution/Lighting Panel Upkeep Policy
 - a. Panel schedules shall be kept in a panel database that is owned by the LTC Electrical Engineer. Electricians assigned to the area will be accountable for the proper labeling of panels in that area.
 - b. Each panel shall have a holder containing two copies of the panel schedule.
 - c. Any electrician who makes a change shall update both copies of the panel schedule by hand and shall properly identify the source information on the utilization equipment which has had the new circuit connected.
 - d. The electrician shall take, mail, or fax an updated schedule to the area's data base editor (this duty can be assigned to the electrician). Editor access can be acquired by contacting the database owner.
 - e. The database editor shall make appropriate changes and shall reprint two copies of the panel schedule and get them back to the electrician assigned to the area.
 - f. The electrician shall place the two new panel schedules in the panel door and destroy the old marked up copies.

L. Location of Isolation Switches

1. For general utilization equipment which requires that lockout/tag out be performed routinely for cleaning or maintenance unrelated to the electrical hazard, a disconnect switch shall be located within sight and accessible. Deviation from this policy may require a QEW to assist with lockout.
2. Utilization equipment which requires lockout/tag out only for reason of electrical hazard, may have a local disconnect installed or be fed from a remote disconnect in compliance with all requirements of the NEC.

M. Warning Signs

1. Warning signs must follow local rules and be written in the local language.
2. All rooms or enclosures that contain exposed live parts operating at greater than 1000 volts shall be posted with a conspicuous sign reading as follows:

DANGER – HIGH VOLTAGE – KEEP OUT

3. All outdoor substations with exposed live overhead circuits/equipment shall be posted as follows: **HARD HAT AREA**

N. Equipment Labelling

Reference: See NFPA 70E

**SAFETY
RELATED WORK
PRACTICES****A. Required Engineering Studies**

Each site that maintains their own distribution system will have the following Engineering studies completed and on file. These studies must be maintained and updated when there is a modification to the system.

1. Fault current analysis
2. Coordination study
3. Arc Flash Analysis.
 - a. Each site should have a completed arc flash analysis as required by NFPA 70E. A qualified electrical engineer from the Engineering Tech Center shall complete the arc flash analysis or review it if completed by an engineering firm
 - b. Each site maintains ownership and the responsibility for updating their arc flash analysis. The Engineering Tech Center shall be used to store and manage the electronic version of the arc flash analysis and will maintain the most current version as provided by each site.
 - c. If an engineering firm is utilized to update an arc flash analysis, the Engineering Tech Center should be contacted to get the most accurate version to update.
 - d. When updates are completed by an engineering firm, the updated electronic version shall be submitted to the Engineering Tech Center so they can document and store the most accurate version.

B. Training

1. QEWs – Crafts that typically require this level of training include:

- Electricians
- High-Voltage Electricians
- Instrument Technicians
- HVAC/Refrigeration Technicians
- Electrical Engineers
- Instrument Engineers

Training	
Type	Title
Self-Study	448188: KELEC1006: SS-Electrical Safety Program
Instructor Led	HS20186: Low Voltage Electrical Safety Training for Affected Employees
Self-Study	HSE IFM: HS02045 QEW Contact Release
OJT	HS02043: QEW Audit – ESWC
One of the following:	
Task Demonstration	HSE IFM: HS02040 QEW Task Demonstration Assessment < 1000V, =HRC0
	HSE IFM: HS02041 QEW Task Demonstration Assessment < 1000V, >HRC0
	HSE IFM: HS02042 QEW Task Demonstration Assessment > 1000V
Trainers only:	
Instructor Led	HSEIFM: HS02044 TTT

Continued on next page

**SAFETY
RELATED WORK
PRACTICES,
CONTINUED**

2. Supervisors of crafts above

Training	
Type	Title
Self-Study	KELEC1006: SS-Electrical Safety Program
Instructor Led	HS20186: Low Voltage Qualified Electrical Safety Training

3. Unqualified Affected Person – Crafts that may typically require this specific training include:

- Maintenance Mechanics
- Industrial Machine Operators
- Production Operators
- Machinists
- Process Automation Engineers/Technicians
- Maintenance Supervisors (not listed in 1. above)

Training	
Type	Title
Self-Study	448188: KELEC1006: SS-Electrical Safety Program
Instructor Led	HS20186: Low Voltage Electrical Safety Training for Affected Employees

4. Facility Operation and Maintenance Management; as well as Facility Engineering and Construction Management

Training	
Type	Title
Self-Study	448188: KELEC1006: SS-Electrical Safety Program

5. Retraining

An employee shall receive additional training (or retraining) according to 70E.

C. "Two Person Rule"

Supervisors must review each energized work task and determine if two people are required for the job.

1. For AC circuits 600 volts and above - At least two qualified people are present while working on or near (within "Restricted Approach" or "Prohibited Approach") exposed energized parts at more than 1000 volts AC.
2. Exception is allowed for the following:
 - a. Routine switching operations required for normal operations and maintenance. (This does not include operations such as breaker racking, fuse changes, etc.)
 - b. Emergency repairs to the extent necessary to safeguard the general public

Continued on next page

**SAFETY
RELATED WORK
PRACTICES,
CONTINUED**

- D. Temporary Safety Grounding (>1000 volts AC)
1. Temporary grounding is to protect personnel working on de-energized equipment. This will protect personnel from the following:
 - a. Induced voltage from adjacent energized lines.
 - b. Induced fault current from adjacent lines.
 - c. Lightning strikes anywhere on the circuit.
 - d. Switching equipment failure or human error.
 - e. Residual voltage stored in de-energized lines or capacitors.
 2. When temporary safety grounds are applicable, the following practices shall be followed:
 - a. The PPE required for the tasks shall be worn until grounds are in place.
 - b. Check for potential before grounding.
 - c. Clamps having adequate current carrying capacity and mechanical strength are used.
 - d. Only approved safety grounding equipment is used.
 - e. Grounds are installed as close to the work area as possible.
 - f. Jumpers are installed grounded end first, then to one of the phases and continue by connecting all phases together. Always connect the grounded conductor to the phase. Removal occurs in the reverse order, removing the connection to ground last.
 - g. A sign is placed on the exterior of all equipment to which grounds have been applied to alert personnel. Signs will not be removed until after the grounds have been removed.
- E. Personal Protective Equipment (PPE)
- All QEWs must:
1. wear *EH rated safety shoes. However, they must NOT wear grounded safety shoes or grounding straps when performing electrical work. It is important not to be grounded when working on electrical equipment or when working on or near energized exposed parts.
 2. wear non-conductive safety glasses. If prescription eye wear is conductive, non-conductive safety glasses must be worn over them.
 3. NOT wear metal jewelry while performing energized work.
- F. Elevated Work Platforms
- As a minimum, activity on an elevated work platform should comply with the Fall Protection Policy. Additional requirements may be implemented for plant site specific activities such as working from a bucket truck on overhead lines.

Continued on next page

**SAFETY
RELATED WORK
PRACTICES,
CONTINUED****G. Lockout/Tagout for Electrical NALO Equipment**

1. Purpose: This procedure establishes the requirements for lockout of electrical energy sources. It is to ensure that conductors and circuit parts are disconnected from sources of electrical energy, locked, and tested before work begins where employees could be exposed to hazardous electrical energy sources. There is no conflict with the company lockout procedure for other mechanical equipment but this procedure recognizes and addresses the unique requirements for the control of electrical energy.

Note: All personnel who are trained (authorized) to perform a Lockout/Tagout operation will adhere to the philosophy of "When In Doubt - Lock It Out". *See USD-SOP-NALO Equip Lockout and Line Breaking Pgm-001 for reference.

2. Preparation
 - a. Review current drawings, tags, labels, and signs to identify and locate all disconnecting means.
 - b. Review other work activity to identify where and how other personnel might be exposed to sources of electrical hazards.
 - c. Establish energy control methods for control of other hazardous energy sources in the area.
 - d. Provide an adequately rated voltage detector to test each phase conductor or circuit part to verify that it is de-energized.
 - e. Where the possibility of induced voltages or stored electrical energy exists, the phase conductors or electrical parts must be grounded before touching them. Where it could be reasonably expected that contact with other exposed energized conductors or circuit parts is possible, ground connecting devices must be applied.

Continued on next page

**SAFETY
RELATED WORK
PRACTICES,
CONTINUED**

3. Individual Employee Control Procedure
The Individual Employee Control procedure can be used when equipment with exposed conductors and circuit parts are de-energized for minor maintenance, and the work is permitted to be performed without the placement of lockout devices on the disconnecting means, provided that the disconnecting means is clearly visible to all employees involved in the work, and the work does not extend beyond the work shift. This procedure applies to plug-in equipment only.
 4. Sequence of Simple Lockout System Procedures
 - a. All personnel performing a Lockout/Tagout operation are familiar with the facility's administrative requirements for this type of operation.
 - b. The affected personnel are notified that a lockout system is going to be implemented and the reason therefore. The qualified person implementing the lockout knows the disconnecting means location for all sources of electrical energy and the location of all sources of stored energy.
 - c. De-energize and disconnect the electrical supply and relieve all stored energy.
 - d. Lockout all disconnecting means with lockout devices.
Note: If no lockout means is available, a Hold card or similar tagout method may be employed. If this is the case, then one additional safety measure must be employed such as opening, blocking, or removing an additional circuit element.
 - e. Attempt to operate the disconnecting means to verify that mechanical operation is prohibited.
 - f. Verify proper instrument operation, then test for absence of voltage.
 - g. Verify proper instrument operation after testing for absence of voltage.
 - h. Where required, install grounding equipment.
 - i. The equipment is now locked out and is in an electrically safe work condition.
 5. Restoring the equipment and/or electrical supply to normal condition.
 - a. Visually verify that the job/task is complete
 - b. Remove all tools, equipment, and unused materials and perform appropriate housekeeping.
 - c. Remove all grounding equipment/conductors/devices.
 - d. Notify all personnel involved that the lockout is complete, that electrical supply is being restored, and to remain clear of the equipment and the electrical supply.
 - e. Perform any quality control tests (Megger®) on the repaired equipment and or electrical supply.
 - f. Remove personal lockout devices (only by the person who installed them).
 - g. Notify the equipment owner that the electrical supply is ready to be returned to normal configuration.
 - h. Return the disconnecting means to their normal condition.
 6. For a simple lockout task, each person installs their own personal lockout device.
-

Continued on next page

**SAFETY
RELATED WORK
PRACTICES,
CONTINUED**

7. Complex Lockout
 - a. A complex lockout is required where one or more of the following exists:
 1. Multiple energy sources
 2. Multiple crews/employees
 3. Multiple crafts
 4. Multiple locations
 5. Multiple employers
 6. Unique disconnecting means
 7. Complex or particular switching sequences; or
 8. Continues for more than one shift, i.e., new workers.
 - b. Each area designates a person in charge who will be involved with a complex lockout/tagout procedure.
 - c. The person in charge develops a written plan of execution and communicates that plan to all persons involved in the task.
 - H. Electrical Manhole Entry
 1. Electrical manhole entry may occur for:
 - Visual inspections
 - Short term observations
 - Long term work
 2. Electrical manhole entry is covered under the Lilly Confined Space Entry program. The confined space entry procedure should be referenced and followed but the following information is pertinent to manhole entries. The confined space entry program rules in the event of a conflict
 3. Entry into manholes does NOT occur when the enclosure contains standing water.
 4. If the construction of an electrical manhole necessitates the use of a free standing ladder for entry and exit, only non-metallic ladders are used.
 5. Each department conducting electrical manhole entries are maintained and kept on file a departmental procedure which includes a required equipment list.
 6. All sources of electrical energy in the manhole above 1000 volts are de-energized prior to work being performed. This work would include pulling in new cable, removing old cable, and performing splices and terminations. Any exceptions must first be approved by electric shop supervision.
 7. If an existing de-energized cable is to be cut, the following apply:
 - a. The cable is tagged in the manhole with the circuit number. The electrician traces the cable from the tag to the location to be cut and clearly mark the cable.
 - b. If the cable has terminations in that manhole with removable test caps, the cable is tested with a voltage sensing device immediately prior to cutting the cable to verify the cable is de-energized.
 - c. If the cable does not have the ability to be tested with a voltage sensing device, all personnel are removed from the manhole when the cut is made. The electrician will obtain proper equipment to remotely cut the cable from outside of the manhole.
-

Continued on next page

**SAFETY RELATED
WORK PRACTICES,
CONTINUED**

8. Circuits are not required to be de-energized for work involving manhole and cable inspections, manhole cleanout, fire wrapping, and drain wire connections. In these instances, the electricians performing the work use their judgment as to whether circuit de-energization is warranted.
 9. The additional responsibilities of the attendant beyond those listed in the confined space entry procedure are:
 - a. Possess a Lilly hand-held radio capable of transmitting and receiving on the site emergency channel.
 - b. Check the area around the electrical manhole and remove any potential hazards.
 - c. Verify that the space around the manhole is clear and free of equipment that might hamper exit from the space.
 - d. Verify that the area around the opening of the manhole is blocked from all traffic, both pedestrian and vehicle. Use barricades, safety cones, marking tape and/or rope.
 10. Training is required for all personnel who enter or attend manholes or who sign the confined space entry permit on a frequency established by HSE.
- I. Electrical Work in Hazardous (Classified) Areas
Note: NALO has no Electrically Hazardous areas. If/when NALO adds an electrically classified area to a facility, the Corporate Electrical Safety Technical Requirements will be added to this document and be adhered to.
- J. Cell Phones
Each NALO site has established cell phone procedures/practices.
- K. Energized Work
Important: **ALL** energized work requires the use of proper PPE, whether an *Energized Electrical Work Permit* is required or not.)
1. Lilly Pre-Job Electrical Safety Form
When working on or near exposed, energized parts > 50 Volts AC/DC use the *Lilly Pre-Job Electrical Safety Form*.
 2. Energized Electrical Work Permit
Except for the following list, energized work is approved by the Director (M3) or above for the area. The *Energized Electrical Work Permit* (available in the Gensuite permitting instance) is used to document this approval.
Exceptions to this requirement are:
 - a. Placing equipment in an electrically safe condition
 - b. Replacement of fuses associated with automation, control, or instrumentation
 - c. Changing light bulbs
 - d. Replacement of plug-in lighting ballasts
 - e. Testing, troubleshooting, voltage measurement
 - f. Current testing with clamp-on meter
 - g. Performing Infrared Scanning or Ultrasonic Testing, if you will not break the restricted approach boundary
 - h. Visual inspections
 - i. Operation of switchgear (e.g. racking breakers in/out or routine switching)

Continued on next page

**SAFETY
RELATED WORK
PRACTICES,
CONTINUED**

3. Energized work shall be performed by qualified personnel
 4. Hazard elimination shall be the first priority in the implementation of safety-related work practices.
- L. Electrical Switching Procedure (Load Center and Substation)
1. A plan of action shall be developed prior to any load center and substation switching. Written procedures shall be completed by plant Electrical Engineer or designated Lilly electricians qualified for the task when required.
 2. The plan shall be communicated to all personnel involved and affected by it.
 3. A review of the corresponding one line diagrams representing the affected equipment shall be completed prior to the development of a written plan of action and to any substation switching.
 4. When load center and substation switching occurs, the device or system affected must be completely checked, as appropriate, for proper and effective lockout. The use of an electrical hold card or a lockout tag may be used in conjunction with the lockout. Where appropriate, ESI (Equipment Specific Instructions) or lockout instructions contained in a Job Plan/MAP shall be used to facilitate the lockout.
 5. All tagging and labeling shall be confirmed to be accurate and any discrepancies must be immediately corrected.
 6. Electrical unit substation and main substation switching is only performed by designated Lilly electricians at sites where they have been trained for the task. At least two electricians must be present, one completing the task and the other observing. During an emergency situation, a trained observer may be used rather than a second electrician.
 7. All electricians performing switching operations are equipped with two-way communication.
 8. Approved PPE must be worn when performing switching operations. Leather protective gloves are worn when using dielectric gloves.
 9. All rubber gloves must be tested or replace every 6 months, rubber blankets must be tested annually and hotsticks must be tested every 2 years.
 10. Before performing work on any circuit, use an approved voltage tester to check for the presence of electricity. The voltage tester should be checked prior to and after each use.
 11. Barricades, cones and/or caution tape are utilized to maintain a safe work zone.
 12. When operating a disconnect or breaker:
 - a. The electrician performing the switching (switching electrician) notifies the electric/high voltage shop/maintenance group and/or site electrical engineer that the switching will be taking place if such switching involves voltages over 1000 volts. If there are no site electrical engineers or high voltage shop, then the facilities manager or designate must be notified.
 - b. The switching electrician notifies the maintenance group/engineer of other utilities if the switching will affect their equipment.
 - c. If the disconnect/breaker is being switched open, the switching electrician and accompanying electrician locks the device according to the lockout and hold card procedures.

Continued on next page

**SAFETY
RELATED WORK
PRACTICES,
CONTINUED**

- d. The switching electrician notifies the electric/high voltage shop/maintenance group and/or the site electrical engineer when the switching is complete
 - M. Electrical Emergency Response Procedures
Reference the Emergency Response procedures for electrical emergency response information.
-

Document Approval Signatures

Approved Date: 13 Mar 2025

Approval Verdict: Approve	Sabry Samuel (C039290@lilly.com) Quality Approver 07-Mar-2025 13:08:52 GMT+0000
Approval Verdict: Approve	Eric Willard (C036806@lilly.com) Approver 07-Mar-2025 13:16:29 GMT+0000
Approval Verdict: Approve	Jen Schultz (L274639@lilly.com) Approver 14-Mar-2025 00:16:53 GMT+0000
Reviewer / Approver Additional Details	N/A